

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Experiment -9

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Subject Name: Advanced Database and Management System

Subject Code: 23CSP-333

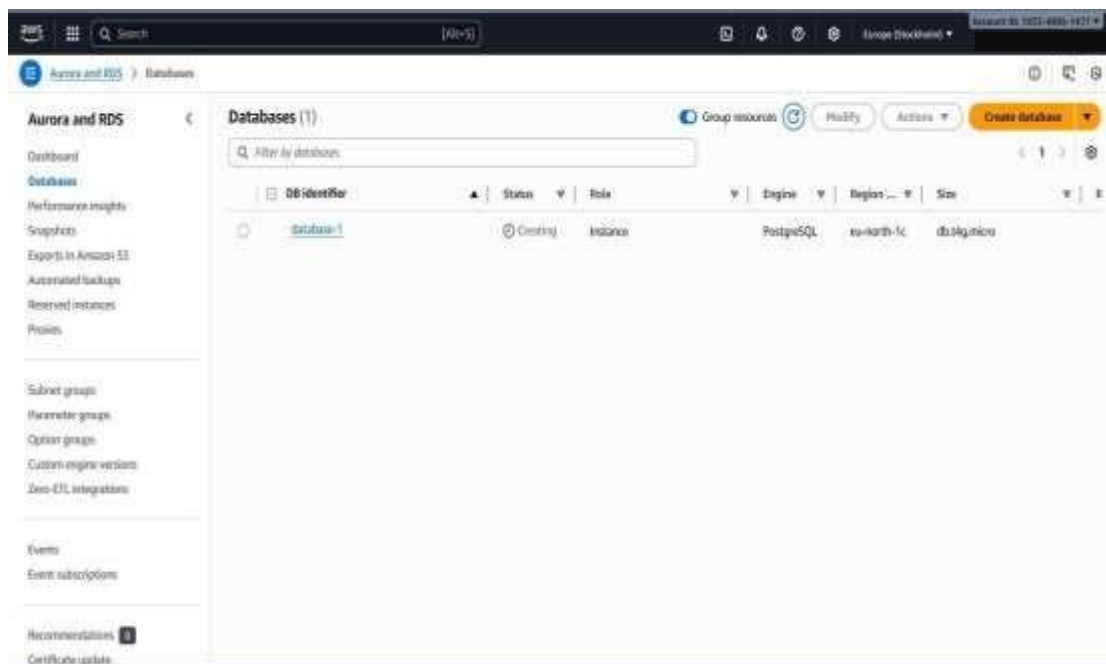
1. Aim

To create, configure, and connect an Amazon RDS PostgreSQL database instance on AWS, and verify successful database operations using a client tool.

2. Objectives:

1. To understand the concept of managed relational database services in AWS (Amazon RDS).
2. To create a PostgreSQL database instance using AWS RDS.
3. To configure DB parameters such as instance size, storage, authentication, and security groups.
4. To connect the RDS instance from a PostgreSQL client (e.g., pgAdmin/psql).
5. To execute basic SQL queries to validate connectivity and database functionality.

3. AWS RDS OPERATIONS AND OUTPUTS -



Amazon AWS

Search

Account ID: 1234-5678-9010

Aurora and RDS

Databases

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations

Events

Event subscriptions

Recommendations

Certificate updates

Databases (1)

Filter by databases

DB identifier

Status

Role

Engine

Region

Size

database-1

Available

Instance

PostgreSQL

eu-north-1c

db.t4g.micro

Create database

CloudShell

Feedback

Console Mobile App

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Security Groups

sg-0805c6095518u040x - default

Edit inbound rules

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Security group rule ID

Type

Protocol

Port range

Source

Description - optional

sg-0c10a5578be47146e

PostgreSQL

TCP

5432

Custom

Q

Delete

sg-063f50f74b4665790

All traffic

all

all

Custom

Q

Delete

Add rule

Cancel

Preview changes

Save rules

CloudShell

Feedback

Console Mobile App

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Instances | Search instances

Instance

Firewall (security group) [info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

Current security group [info](#)

Select security group

default - sg-05f5b89551b0d4a1c [info](#)

SGP: sg-05f5b89551b0d4a1c

☒ Compare security group rules

Configure storage [info](#) [Advanced](#)

Size: 30 GB Type: gp3 Root volume, 3000 IOPS, fast encrypted

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance.

☒ Click refresh to view backup information

Use tags that you assign determine whether the instance will be backed up by any (Data Lifecycle Manager) policies.

0 x File systems [info](#)

Summary

Number of instances [info](#)

1

Software image (AMI)

Microsoft Windows Server 2019 - [read more](#)

ami-0b1b8d51114ba1a

Virtual server type (instance type)

t3.micro

Firewall (security group)

default

Storage (volumes)

1 volume(s) - 30 GB

[Previous steps](#)

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Aurora and RDS | **database-1** [Modify](#) [Actions](#)

Summary

DB identifier: database-1

Status: Available

Class: db.t4g.micro

Role: Instance

Engine: PostgreSQL

Current activity:

Region & AZ: us-east-1c

Recommendations

Connectivity & security | **Monitoring** | **Logs & events** | **Configuration** | **Zero-ETL integrations** | **Maintenance & backups** | **Data ml**

Connectivity & security

Endpoint & port

Endpoint: database-1.cx2gncglgert7au.cxst-h-1.cx1amzsmnwn.com

Port: 5432

Networking

Availability Zone: us-east-1c

VPC: vpc-0f8b8fa3d0b9c74d5a

Subnet group: default-vgp-02f8dc3a3d5fc74b0a

Subnets: subnet-05a3387618171d1f9, subnet-0f82a207fc34633f

Security

VPC security groups: default-vgp-05f5b89551b0d4a1c

Active:

Publicly accessible: No

Certificate authority: [info](#)

db-ca-rs2046-g1

Certificate authority date: May 25, 2021, 01:29 (UTC+01:00)

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Create database

Free plan has access to limited features and resources

The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. [Learn more](#)

Choose a database creation method

☐ **Standard create**

Use set of all the configuration options, including need for availability, security, backups, and maintenance.

☒ **Easy create**

Use our recommended best-practice configurations. Some configuration options can be changed after the database is created.

Configuration

Engine type [info](#)

☒ **Aurora PostgreSQL** [info](#)

☒ **Aurora PostgreSQL** [info](#)

☐ **MySQL**

☒ **PostgreSQL**

☐ **MySQL**

☐ **Oracle**

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Publicly accessible	No	Yes
Database port	5432	Yes
DB instance identifier	database-1	Yes
DB engine version	17.6	Yes
DB parameter group	default.postgres17	Yes
Database Insights -		

The screenshot shows the AWS Management Console for an Amazon RDS instance. The left sidebar lists navigation options like 'Database', 'Performance insights', 'Graphs', etc. The main panel is titled 'Connectivity & security' and is divided into three sections:

- Endpoint & port:** Shows the endpoint as 'database-1.c2gqglpyn7aw-north-1.1.ck.amazonaws.com' and the port as '5432'.
- Networking:** Shows the availability zone as 'eu-north-1c', the VPC as 'vpc-f4f86d3c2d8fc7d07e', the subnet group as 'default-vpc-0d78b3e2d8fc744da', and the subnets as 'subnet-f33a32397f011929', 'subnet-0f5dbac027f158333f', and 'subnet-0f27f15ac532a56a45'. The network type is 'IPv4'.
- Security:** Shows the VPC security groups as 'default (sg-5b53a85315b384a2)' and 'Active'. It also shows 'Publicly accessible' as 'No', 'Certificate authority' as 'info', 'Certificate authority date' as 'May 25, 2001, 05:20 (UTC+05:30)', and 'DB instance certificate expiration date' as 'November 09, 2026, 17:38 (UTC+05:30)'.

The screenshot shows a PostgreSQL client application window with the 'Register - Server' dialog box open. The dialog has tabs for 'General', 'Parameters', 'SSH Tunnel', 'Advanced', 'Post Connection SQL', and 'Tags'. The 'General' tab is active, showing fields for 'Port' (5432), 'Maintenance database' (postgres), 'Username' (postgres), 'Password' (postgres), and 'Save password?' (checked). A red error message at the bottom of the dialog reads: 'Unable to connect to server: connection timeout expired'. In the background, another window shows a list of tools including 'Create PDF', 'Combine Files', 'Edit PDF', etc.

EC2

Instances

Dashboard

AWS Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Instances (1)

1 min updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Find instance by attribute or tag (case-sensitive)

All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 ID
☐	myserver	i-00d5c5b63efed746b6	Running	t3.micro	Initiating	View alarms	eu-north-1a	ec2-51-20-18

Select an instance